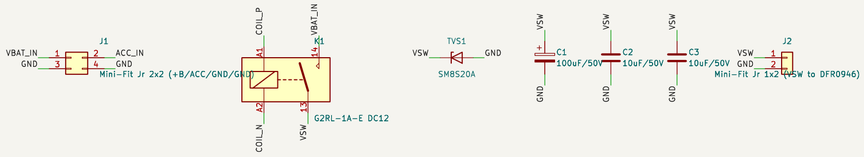
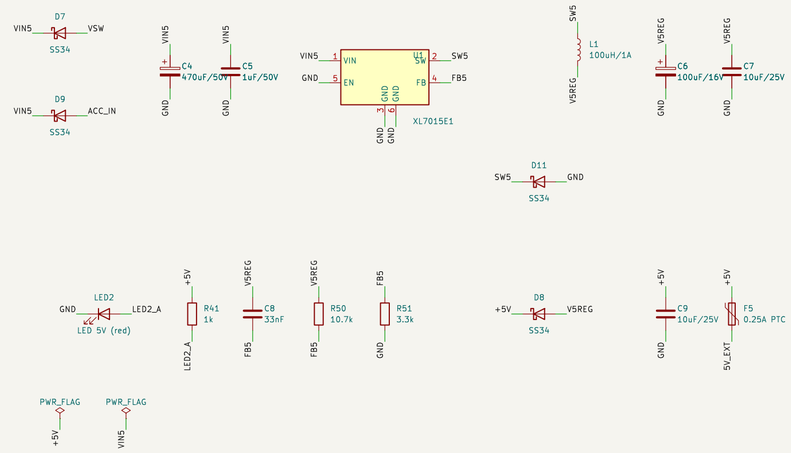
1. POWER INPUT / MAIN SWITCH (7 A path: J1 -> K1 -> J2)



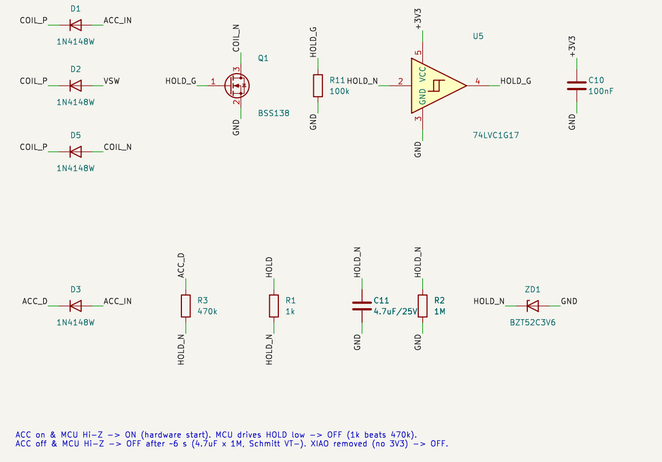
K1 contact 13–14 carries up to 7 A (PC 60 W at 9 V crank). F1 7.5 A / F2 1 A blade fuses are in the harness.
TVS1 after relay: zero leakage when off. Clamp 32.4 V < DFR0946 35 V max.

2. 5 V AUX SUPPLY (XL7015, Vout 5.30 V)



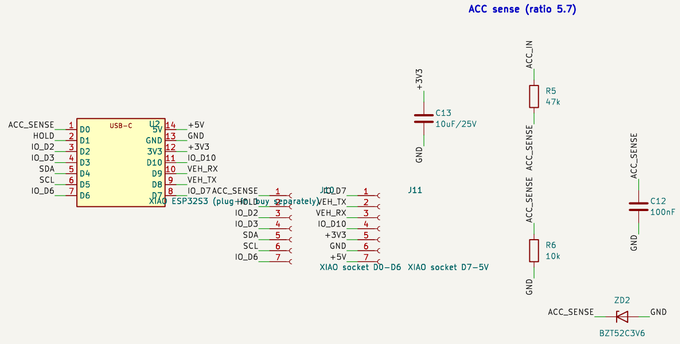
Vout = 1.25 x (1 + 10.7k/3.3k) = 5.30 V. After D8: 4.75–5.1 V to XIAO 5V pin.
Keep C5–U1–D11–L1–C6 loop tight. EN tied to GND (enabled).

3. RELAY HOLD (ACC hold + MCU override + Schmitt)



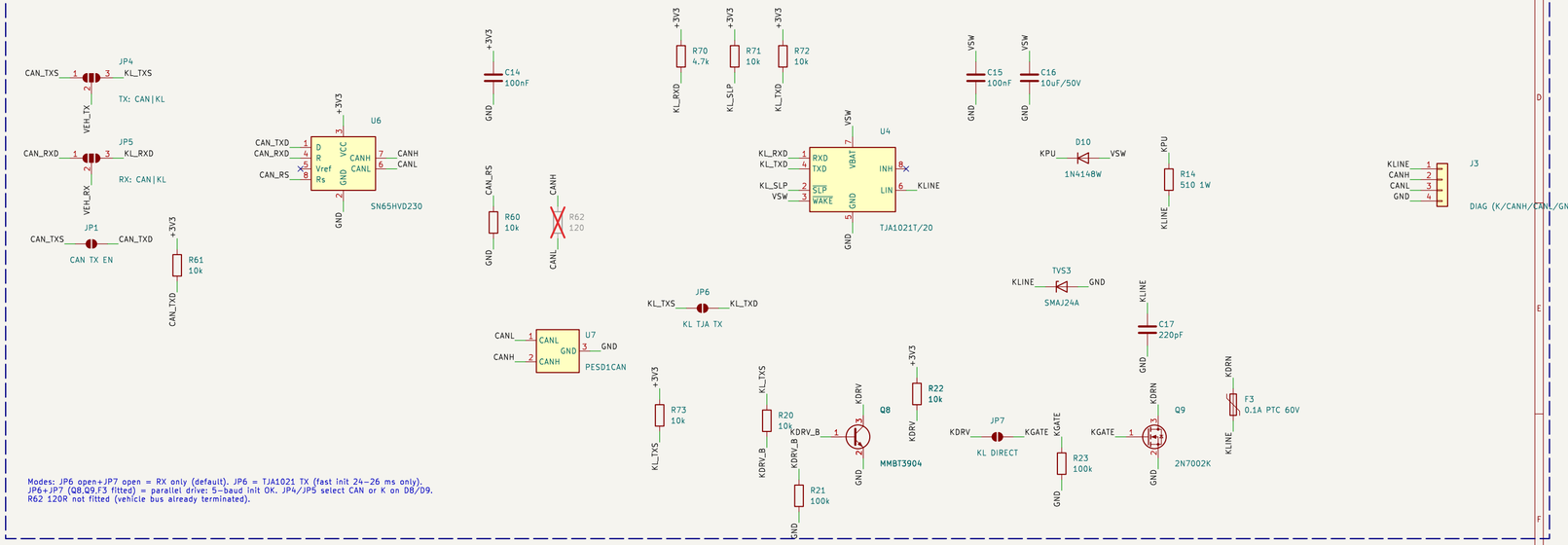
ACC on & MCU Hi-Z -> ON (hardware start). MCU drives HOLD low -> OFF (1k beats 470k).
ACC off & MCU Hi-Z -> OFF after ~6 s (4.7uF x 1M, Schmitt VT-). XIAO removed (no 3V3) -> OFF.

4. MCU: Seeed XIAO ESP32S3 (socketed)



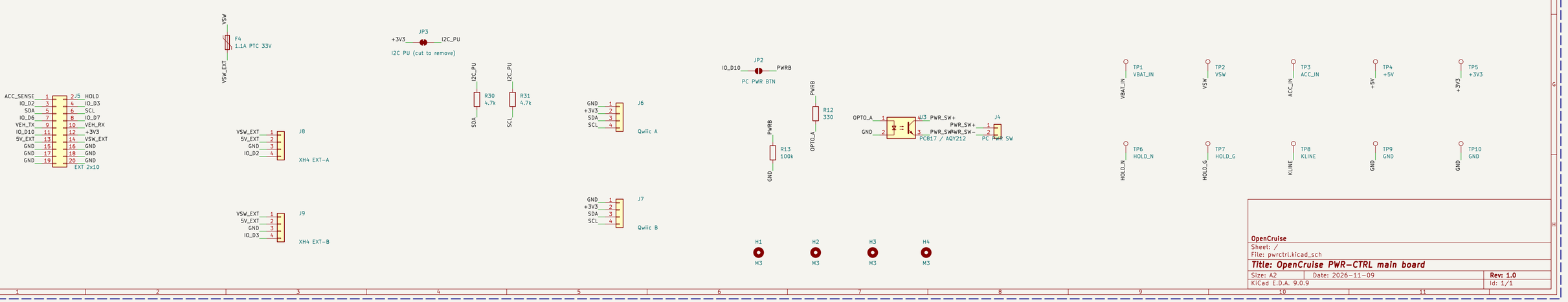
D6/D7 (GPIO43/44) carry UART0 boot log: avoid as outputs needing a quiet boot.
PC USB goes to XIAO USB-C through the separate VBUS-block adapter.

5. VEHICLE BUS: CAN (SN65HVD230) / K-LINE (TJA1021) – default RX-only



Modes: JP6 open+JP7 open = RX only (default). JP6 = TJA1021 TX (fast init 24–26 ms only).
JP6+JP7 (Q8,Q9,F3 fitted) = parallel drive: 5-baud init OK. JP4/JP5 select CAN or K on D8/D9.
R62 120R not fitted (vehicle bus already terminated).

6. EXPANSION / I2C / PC POWER BUTTON / TEST POINTS / MECHANICAL



OpenCruise	
Sheet: /	
File: pwrctrl.kicad_sch	
Title: OpenCruise PWR-CTRL main board	
Size: A2 | Date: 2026-11-09	**Rev: 1.0**
KiCad E.D.A. 9.0.9	Id: 1/1